

AI · PREDICTIVE MAINTENANCE · STEEL MANUFACTURING

SHERLOCK AI

The Intelligent Maintenance Wizard for Steel Manufacturing

WHAT IT DOES

An AI-powered platform that consolidates operational data, maintenance records, sensor streams and engineer queries into explainable diagnostics, failure prediction and prioritized action — with a live, queryable 3D digital twin.

DOCUMENT

Project Overview & Architecture 1.0

VERSION

1.0

PLATFORM

Autonomous Plant Intelligence

DECISION-SUPPORT FOR RELIABILITY TEAMS



The Problem

Costly downtime

Steel plants face significant losses from unplanned equipment downtime

Fragmented knowledge

Teams rely on scattered SOPs, manuals, logs, sensor alerts and expertise

Manual & reactive

Root-cause analysis and planning are manual, slow and after-the-fact

No live visibility

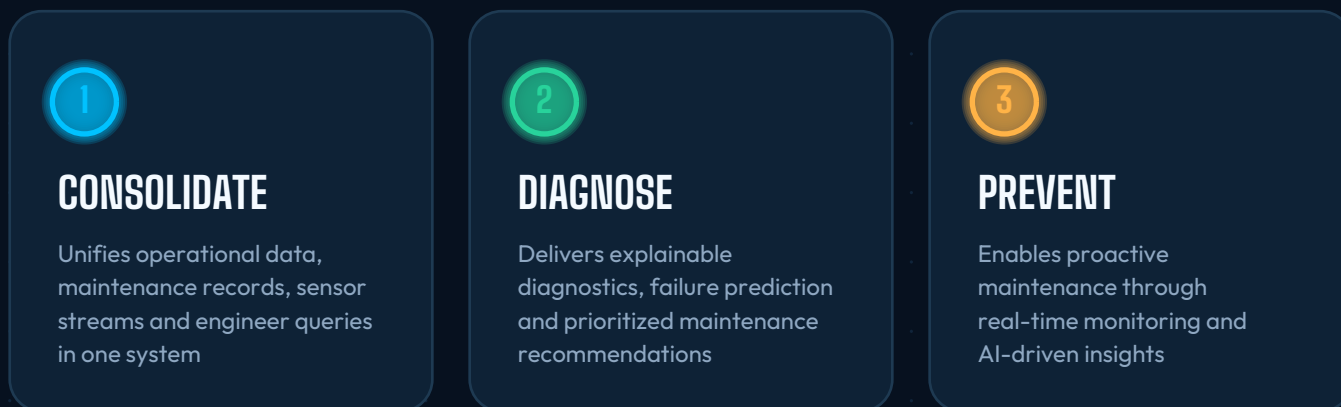
Limited real-time insight drives safety risk, production loss and higher cost

THE GAP

No single, context-aware system consolidates these sources, reasons over them, and recommends prioritized action.

The Solution

An AI-powered predictive maintenance and plant intelligence platform.



FROM DATA TO ACTION



OUTCOME

A shift from reactive firefighting to proactive, explainable, data-driven maintenance.

Key Features



01

Root-Cause Diagnosis

AI diagnosis with confidence scores



02

Failure & RUL Prediction

Failure probability + remaining useful life



03

Real-Time Alerting

Live abnormality detection & alerts



04

3D Digital Twin

Interactive twin of the steel plant



05

AI Assistant

Natural-language helper for engineers



06

Smart Prioritization

Criticality · spares · procurement lead time



07

Operations Dashboards

Safety, energy & production monitoring



08

Automated Reporting

Executive & maintenance PDF reports

Core Innovation



LIVE 3D DIGITAL TWIN

A queryable, real-time 3D twin of the plant fused with AI-driven maintenance intelligence



MULTI-AGENT ARCHITECTURE

A LangGraph supervisor routing to domain-specific reasoning agents



EXPLAINABLE AI

Recommendations linked directly to sensor evidence and operational data

01

System Architecture

PRESENTATION LAYER

Command Center · 3D Digital Twin · AI Command Assistant · Predictive Reliability · Safety & Risk Portal · Executive Reports



SERVICE LAYER

Auth · Dashboard · Maintenance · Safety · Energy · Production · Reporting · AI Chat services



INTELLIGENCE LAYER

LangGraph Supervisor → Maintenance · Safety · Energy · Production · Reporting agents · Gemini reasoning engine



REAL-TIME ENGINE

Live sensor monitoring · failure-probability calculation · risk classification · RUL prediction · alert generation



DATA LAYER

Maintenance records · equipment logs · production · energy · safety · spare-parts inventory · future RAG knowledge base

Technology Stack

FRONTEND

- Next.js 14
- React 18
- TypeScript
- Tailwind CSS
- Three.js
- Recharts
- Axios
- Lucide React
- Date-fns

BACKEND

- FastAPI
- Uvicorn
- Python 3.11+
- JWT Auth
- Passlib + Bcrypt
- Pydantic Settings
- Python Dotenv
- CORS Middleware
- GZip Compression

AI & MACHINE LEARNING

- LangGraph
- Gemini 2.5 Flash
- OpenRouter API
- Multi-Agent AI
- Explainable AI
- Failure Prediction

DATA PROCESSING

- Pandas
- NumPy
- CSV Data Sources

REPORTING

- ReportLab
- PyPDF

DATABASE

- PostgreSQL (Ready)
- SQLAlchemy
- AsyncPG

DEVOPS & DEPLOYMENT

[Docker](#)[Docker Compose](#)[Vercel - Frontend](#)[Render - Backend](#)

Benefits



Reduced downtime

Catch failures before they stop the line



Faster diagnosis

Pinpoint faults in minutes, not hours



Better planning

Maintenance scheduled around real risk



Enhanced safety

Equipment risk and HSE in one view



Higher utilization

Assets run longer, smarter



Lower cost

Fewer emergencies, leaner spares



Operational efficiency

Proactive, not reactive operations



Data-driven decisions

Every call backed by evidence

Future Enhancements

01 RAG knowledge retrieval

Reasoning over manuals & SOPs with citations

02 Feedback-driven learning

Improve from engineer actions & outcomes

03 Domain-specific LLM

Fine-tuned model for steel maintenance

04 Industrial IoT

MQTT / OPC-UA live equipment integration

05 Digital maintenance logbook

Automatic per-asset activity log

06 Advanced forecasting

Deeper predictive analytics & trends

From reactive to predictive.

Intelligent maintenance decision-support for steel-plant reliability teams.

AT A GLANCE

Live 3D digital twin

LangGraph multi-agent AI

Explainable diagnosis

Failure & RUL prediction

Real-time alerting

Prioritized action

GET STARTED

`docker-compose up --build`

API docs at `/api/docs`